

left side of the Explorer window. Place two resistors from the lumped components and a DC source from sources. Connect them using the wire (*Ctrl+E*) ; now place the ground symbol near the negative of the cell. (Specifying a ground point is very essential in all circuit simulators.)

Select *Insert->Wire* label and click on the point between the two resistors and name it *Vout* (or anything you want). Now, from simulations, place the DC simulation block; the schematic should look like Figure 2. Now save it and go to *Simulation ->Simulate*. A new work area appears, and also the Explorer window changes to the diagram tab. From there, select a table and drag it to the work area. Choose the tabular (list of values) diagram and place it on the data display page. After dropping the tabular, the diagram dialog appears. By double clicking the *Vout.V*, the graph (i.e., values in a tabular plot) is added to the diagram (Figure 3). Besides the node voltage *Vout.V*, the current through the DC voltage source *VI.1* is also available. Only items listed in the dataset list can be put into the graph.

Available dataset items

Depending on the type of simulation you performed, you will find the following types of items in the dataset:

1. *node.V*—DC voltage at node
2. *name.I*—DC current through component name
3. *node.v*—AC voltage at node
4. *name.i*—AC current through component name
5. *node.vn*—AC noise voltage at node
6. *name.in*—AC noise current through component name
7. *node.Vt*—transient voltage at node
8. *name.It*—transient current through component name
9. *S[1,1]*—S-parameter value

Depending on the type of the graph, you have various options to choose for it. For a tabular graph, there is number precision, as well as type of number notation (important for complex values). Press the 'Ok' button to close the dialogue box. You will get something like what's shown in Figure 4. You can change the properties of the components by double-clicking on them and changing the properties in the Property box.

Parameter sweep

Now let's look at how to plot the forward characteristics of a p-n junction diode. The diode can be obtained from the non-linear components. Since we are going to plot how voltage varies as current passing through the diode is changed, let's use a DC current source, and change its numerical value to a variable, *If*. This can be done by double-clicking the component and editing the component dialogue box. Then wire up the circuit (don't forget to ground it), and place a wire label at a point between the diode and the positive of the current source. Name it *Vd*.

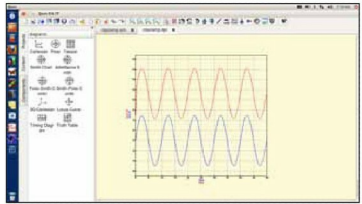


Figure 4: A graphical output

Now place the DC simulation block and also place the parameter sweep block. In the parameter sweep properties, give the name of the DC simulation block you placed as *DC1*, the parameter as *If* (the parameter you wish to change), and then give the type of variation you want—whether linear or logarithmic. Then, give the starting and ending values (I gave 0 to 10). Also give the number of points you need (100 will give you a pretty neat graph). Now simulate, and in the data display page, place the Cartesian diagram instead of the table as in the previous example, and choose the parameter you want to display. You will have the forward characteristics of the diode, with voltage on the y axis and current along the x axis.

What next?

I believe that students should be taught such simulation software along with their lab experiments, and also that students should try to use such tools when doing their projects. In this way, a lot of time and money can be saved. You can analyse how a system will respond in a given situation.

Well, I hope this information is enough to get you started. There are lots of tutorials available on the Net. There is a lot more that you can do with QUCS, like digital simulation, harmonic analysis, transient analysis, etc. If you really liked the software, try contributing back—after all, FOSS is all about sharing, giving and learning. 

References

- [1] <http://qucs.sourceforge.net/>
- [2] http://en.wikipedia.org/wiki/Quite_Universal_Circuit_Simulator

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